

OneSolar

One Prompt. One Design. One Solar.

Designed and developed by a handful of engineers

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1. About OneSolar

OneSolar is an AI-native solar design and engineering platform built to bring the same shift to solar photovoltaic (PV) design that AI coding agents have brought to software development. Today, a developer can describe a feature in plain English and have an AI agent write, test, and refine the underlying code without the developer needing to write it line by line. OneSolar applies this same principle to solar engineering: a solar designer, installer, or EPC (Engineering, Procurement and Construction) firm describes a site and a requirement in plain language, and OneSolar produces a complete, engineering-grade solar design in return — including the panel layout, shading and energy yield simulation, electrical and code compliance checks, bill of quantities, and a client-ready proposal.

The company is built around a single core belief: solar design should not require the person doing it to already be an expert in specialised solar engineering software. Just as natural-language prompting has replaced a large share of manual coding work, OneSolar is built to replace manual, click-heavy panel placement and simulation work with a conversational, iterative design workflow, while keeping the engineering underneath rigorous, standards-compliant, and reliable enough to be used for real client proposals and financing.

2. The Problem We Are Solving

Solar design today, particularly across the Indian residential and commercial rooftop segment, is slow, manual, and heavily dependent on specialised design software that takes real time and training to use well. Designers spend hours placing panels, adjusting layouts around shading and roof obstructions, recalculating string and inverter sizing, and manually assembling proposals for every single site — even though a large share of this work follows patterns that are repeatable and well understood by an experienced designer.

This creates a direct bottleneck on how quickly a solar company can turn a sales lead into a signed, installable design. Every hour spent on manual layout and recalculation is an hour not spent on the next customer, and smaller installers in particular often cannot justify the cost or learning curve of the more powerful enterprise design tools built for large utility-scale developers.

Existing design software has begun adding AI-assisted features, but the majority of these remain limited to guided automation layered on top of a traditional point-and-click interface. They speed up individual steps, but the person operating the software still needs to understand the software itself. OneSolar is built to go a meaningful step further: a true conversational design agent that takes a natural-language brief and iterates on the complete design end-to-end, the way an experienced senior designer would if simply instructed by a colleague in plain language.

3. What OneSolar Does

3.1 Natural-Language Design

A user describes the site and the requirement in plain English or in Hindi and Hinglish — for example, a rooftop capacity target, a carport layout, or a specific constraint such as avoiding a skylight, staircase, or water tank — and OneSolar generates a first-draft design directly from that description, using satellite imagery and site data to build an accurate model of the site.

3.2 Engineering-Grade Simulation

Every design produced by the platform is backed by genuine simulation rather than approximation. This includes shading analysis, hour-by-hour energy yield estimation, and string and inverter sizing calculated against real site conditions and equipment parameters, so that the resulting output is suitable for client proposals, financing conversations, and permitting submissions, and not simply an illustrative sketch of what a system might look like.

3.3 Compliance and Standards

Designs are checked against applicable Indian electrical and safety codes and relevant DISCOM (Distribution Company) requirements, and are structured to align with government schemes such as PM Surya Ghar Muft Bijli Yojana, so that residential systems remain eligible for the subsidy pathways available to homeowners.

3.4 Iterative Refinement Through Conversation

Once a first draft exists, changes are made the same way a developer refines code together with an AI coding agent — by describing the change in plain language. The system updates the layout, re-runs the relevant simulation, and reflects the change immediately, removing the need to manually rebuild the design from scratch every time a requirement changes.

3.5 Proposal and Documentation Output

OneSolar also produces the full set of downstream documentation a completed design normally requires, including single-line electrical diagrams, a bill of quantities, financial and payback analysis, and a branded, client-ready proposal document — closing the gap between a finished design and a signed customer.

4. Who OneSolar Is Built For

OneSolar is designed primarily for the people who currently spend their working hours inside solar design software, and for the businesses that depend on their output:

- Residential and commercial rooftop solar installers who need to turn around accurate quotes and designs quickly and at volume.
- EPC firms and in-house design teams looking to reduce design turnaround time without adding headcount.

- Growing solar businesses that cannot yet justify the cost of enterprise-grade, utility-scale design software.
- Sales and business development teams who need a fast, credible design to close a deal during or immediately after a site visit.

5. Why Now

India is targeting a substantial expansion of its renewable energy capacity over the coming years, with solar as the primary driver, and this growth is being actively supported by continuing government subsidy programmes for rooftop adoption. At the same time, the underlying AI technology required to build a genuinely conversational, agentic design tool — language models capable of reliable multi-step reasoning and reliable use of external tools — has only recently matured to a point where a product like OneSolar becomes realistically possible to build well. This combination of a fast-growing market and newly available technical capability creates a timely opportunity to build a design platform that is native to this new way of working, rather than one that simply adds AI features onto older, click-based software built for a previous generation of tools.

6. Our Approach

OneSolar is built on the principle that speed should never come at the cost of engineering accuracy. Rather than using AI to guess at a plausible-looking layout, the platform uses AI to orchestrate real, validated engineering tools and calculation methods, treating simulation, compliance checking, and applicable standards as underlying systems that the AI agent calls upon and reasons over, rather than steps it simply approximates from pattern recognition. The natural-language interface is the experience layer that the user interacts with; the engineering underneath it remains rigorous, transparent, and verifiable at every step.

7. Current Stage

OneSolar is currently at an early, pre-product stage. The immediate focus is on validating the core problem directly with solar installers and EPC firms, understanding their existing workflows and pain points in detail through direct conversation, and using that input to shape the first working version of the platform before committing to full-scale development. This approach is deliberate: the goal is to build a product that is shaped by real, first-hand feedback from the people who will use it every day, rather than one built on assumption alone.

8. Industry Pain Points and Projected Impact: The Evidence Base

The case for OneSolar does not rest on assumption. It rests on a documented, citable pattern: solar deployment is accelerating faster than the skilled workforce and design tooling available to support it, manual design and rework carry a measurable cost across the construction and solar sector, and comparable AI-assisted design tools already in the market have published verifiable efficiency gains. This section lays out that evidence, stage by stage, with sources listed in Section 9.

8.1 The Market Is Growing Faster Than the Workforce That Serves It

India's rooftop solar sector is in the middle of its fastest expansion to date. Cumulative grid-connected rooftop solar capacity grew from roughly 2.5 gigawatts in 2018 to approximately 16.9 gigawatts in 2024, and reached 30.11 gigawatts by May 2026, driven substantially by the PM Surya Ghar: Muft Bijli Yojana scheme (Ref. 1, 2, 3). Under that scheme alone, installations have accelerated from around 5,038 households per day in October 2025 to nearly 16,328 households per day in July 2026, with the programme surpassing 5 million beneficiary households against a target of 10 million by FY2026-27 (Ref. 4). Rooftop installations grew 125% year-over-year in the first quarter of 2026 alone (Ref. 5).

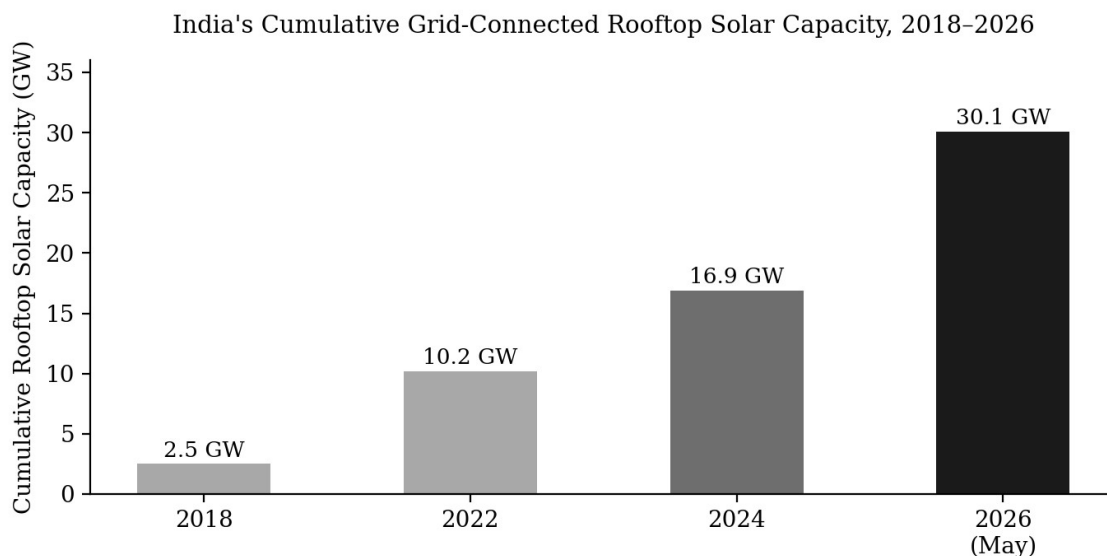


Figure 1: India's cumulative grid-connected rooftop solar capacity, 2018–2026. Source: MNRE data as reported by GreentechLead and 500 Seconds (Ref. 1, 3).

This pace of growth is running directly into a documented shortage of skilled design and engineering talent. Mercom Capital Group's CEO has publicly estimated that the shortage of skilled workforce is slowing India's clean energy industry growth by 10 to 20 percent, directly limiting companies' ability to execute and scale projects (Ref. 6). Globally, the IEA's World Energy Employment 2025 report found solar PV employment surpassed 5 million people in 2024, with distributed (rooftop) solar accounting for two-thirds of all solar employment despite representing less than half of installed capacity — meaning distributed solar is disproportionately labour-intensive per megawatt deployed (Ref. 7). IRENA and the ILO's 2025 Renewable Energy and Jobs review puts India's renewable energy workforce at 1.28 million as of 2024, a workforce that has to stretch further every year as installation volumes climb (Ref. 8). This is precisely the gap a natural-language design agent is positioned to close: it does not require the market to produce more trained CAD-literate designers at the same rate solar capacity is growing — it lets the designers who already exist do more with the time they have.

8.2 Manual Design and Rework Carry a Measurable, Published Cost

The inefficiency in manual solar and construction design is not anecdotal — it has been measured repeatedly across the broader construction industry, of which solar EPC is a part. A Dodge Data & Analytics study found that rework and delays cost the United States construction industry approximately \$177 billion annually, equivalent to roughly 5 percent of total construction spending, and that change orders average 10 percent of contract value, rising as high as 25 percent on complex projects (Ref. 9). The Construction Industry Institute and related Navigant Construction Forum research places rework at a median of roughly 5 percent of project cost across multiple studies spanning more than three decades (Ref. 10). Design-related errors and omissions are consistently identified as the dominant driver: one industry analysis attributes 56.5 percent of construction cost overruns and 70 percent of project rework directly to design changes and design inconsistencies (Ref. 9).

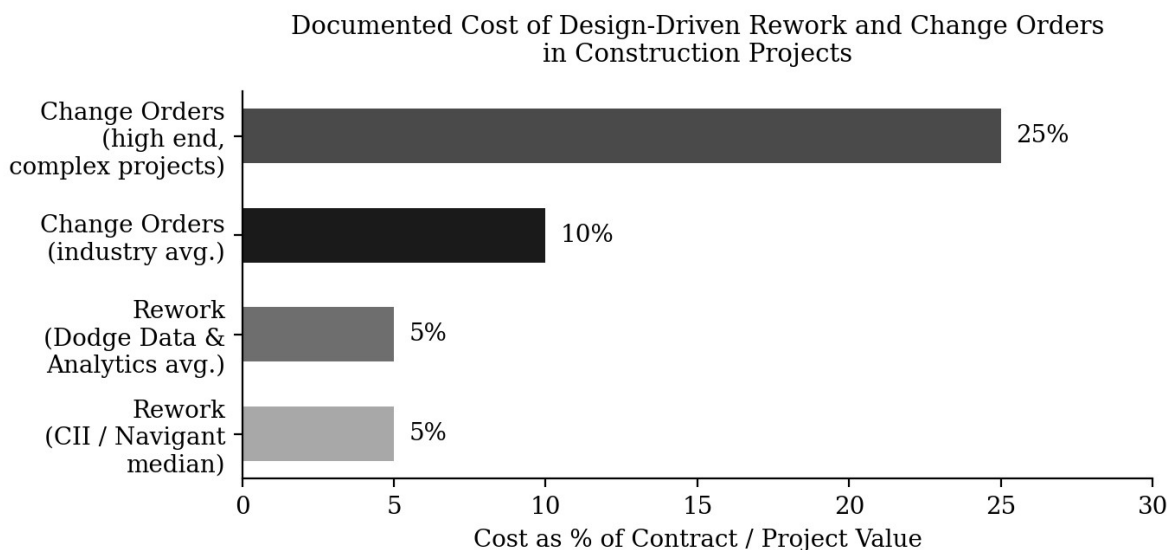


Figure 2: Documented cost of design-driven rework and change orders as a percentage of project value, across multiple industry studies. Sources: Dodge Data & Analytics via Rhumbix; Construction Industry Institute / Navigant Construction Forum (Ref. 9, 10).

Solar-specific evidence points the same direction. Aurora Solar — the largest residential solar design platform in the United States — has published customer case studies showing that AI-assisted design cuts change orders from 40 percent of projects to 10 percent, saves 48 to 72 hours per project through eliminated duplicate work, and reduces plan-set delivery time by 7-fold, while increasing individual designer capacity by 3-fold (Ref. 11, 12). CertainTeed, a Saint-Gobain subsidiary, reported that AI-assisted design compressed a workflow that previously took 3 to 4 weeks — from initial site information to a finalised, revised design — into a single 10-minute session conducted live with the customer (Ref. 13).

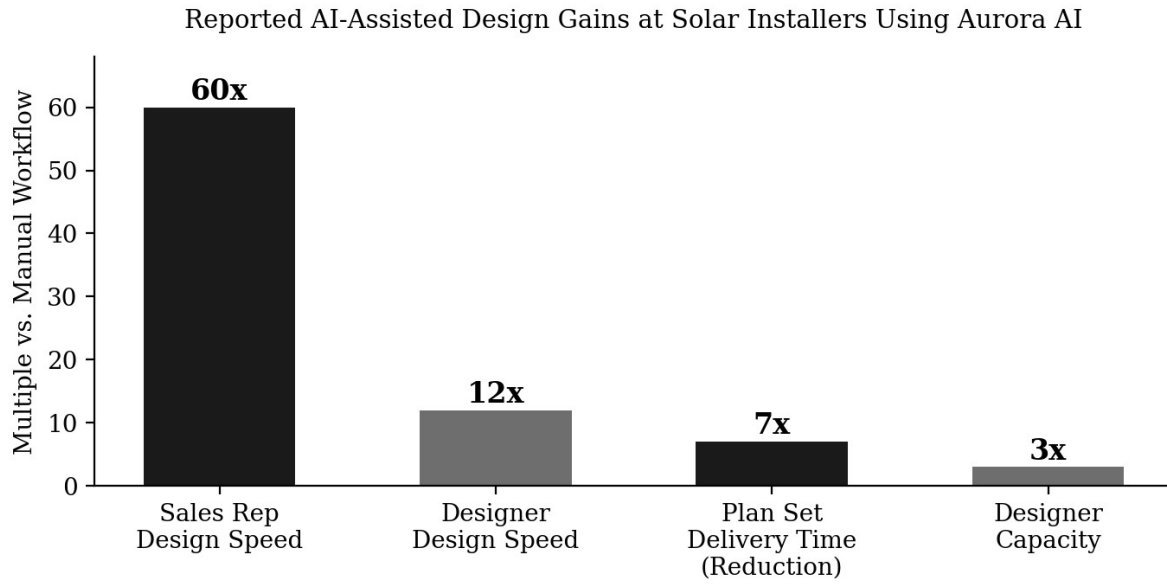


Figure 3: Reported AI-assisted design efficiency gains published in Aurora Solar customer case studies. Source: Aurora Solar, “Make the Switch” (Ref. 11).

These are not projections from an unproven concept — they are outcomes already achieved and published by the market's largest incumbent using a guided-automation approach. OneSolar's premise is that a genuinely conversational, agentic layer — rather than a guided point-and-click AI feature — can extend these same categories of gains further, particularly for the smaller and mid-sized Indian installers who cannot access or afford enterprise platforms built primarily for the US market.

8.3 A Growing Software Category Confirms Market Readiness

Independent market research also confirms that capital and demand are actively flowing into this category. The global solar design software market was valued at approximately \$1.24 billion in 2024 and is projected to grow at a compound annual growth rate of roughly 13.2 percent through 2033, reaching an estimated \$3.4 billion, with the Asia-Pacific region — including India — identified as the fastest-emerging growth engine due to expanding solar targets and rising installation volumes (Ref. 14).

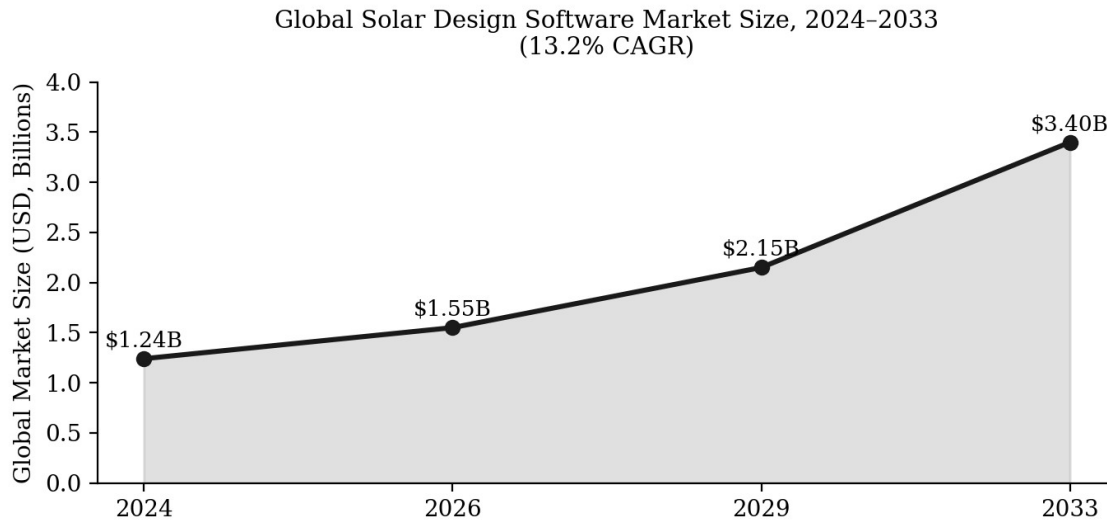


Figure 4: Global solar design software market size and growth trajectory, 2024–2033. Source: Dataintelo, Solar Design Software Market Research Report (Ref. 14).

This confirms the category itself is not speculative: buyers are already paying for solar design software, that spend is growing faster than the solar market overall, and the shift toward AI-native tooling within that category is accelerating rather than plateauing.

8.4 Projected Impact of OneSolar

Drawing on the documented outcomes above — rather than internal measurement, given the platform's current pre-product stage — OneSolar's design targets are grounded in what comparable AI-assisted tools have already demonstrated in the market: a meaningful reduction in time spent per design stage, a multi-fold increase in the number of designs a single designer can complete per day, and a corresponding reduction in costly downstream rework and change orders. If OneSolar can replicate even a conservative fraction of the gains Aurora Solar and Enerflo customers have already published — using a more natural, prompt-driven interface aimed specifically at the Indian rooftop and small-to-mid EPC segment — the effect compounds into three outcomes for the businesses that adopt it: faster conversion of leads into signed, installable designs; fewer design-driven change orders and the associated cost and schedule risk; and access to bankable design capability for installers who today cannot justify enterprise software costs. Validating these targets against real workflows at partner installers, as outlined in Section 7, is the immediate next step before any of these figures are treated as confirmed outcomes rather than evidence-based targets.

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